

BANGALURU INTL AP, IndiaWMO: **427056**Lat: **13.200N**Lon: **77.700E**Elev: **915**StdP: **90.81**Time Zone: **5.50 (IND)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	13.0	-1.0	3.9	29.3	1.9	4.8	28.8	6.9	25.6	6.3	25.5	0.7	100	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.5	37.2	20.5	35.8	20.7	34.2	20.2	26.6	28.9	26.0	28.4	25.4	28.2	3.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	24.1	27.3	25.2	22.8	26.8	25.0	22.5	26.6	89.4	29.4	87.2	27.6	84.6	28.5	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.6	9.8	36.6	1.7	2.1	8.5	38.2	7.6	39.4	6.6	40.6	5.4	42.2		
(5)			9.1	25.8	1.7	2.0	7.9	27.2	6.9	28.4	6.0	29.5	4.7	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	21.4	23.3	26.4	28.5	28.0	25.9	25.2	24.8	24.9	24.3	22.9	21.8
	DBStd	2.91	2.14	2.36	2.34	2.19	1.85	1.84	1.74	1.74	1.67	1.83	1.89	2.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	5392	354	371	507	556	557	478	470	458	446	443	387	366
	CDD18.3	2351	97	138	249	306	298	228	212	199	196	185	137	108
	CDH23.3	21153	897	1593	3056	3763	3230	1874	1442	1248	1349	1246	783	672
(14) Wind	CDH26.7	8568	225	678	1602	2089	1572	651	409	322	373	345	167	135
	WSAvg	3.5	2.2	2.5	2.7	2.7	3.6	5.8	6.1	5.1	4.1	2.5	2.4	2.5
	PrecAvg	963	6	5	12	32	81	88	115	141	144	187	112	41
	PrecMax	1412	47	61	105	104	250	265	211	284	228	505	435	167
	PrecMin	678	0	0	0	2	9	30	34	45	51	20	4	0
	PrecStd	197	10	12	20	23	52	51	47	56	49	90	90	39
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	35.2	38.0	39.8	37.2	34.2	33.2	33.1	33.1	33.1	31.1	31.6
		MCWB	19.9	18.9	20.3	21.5	22.3	24.2	23.9	24.6	24.5	22.6	21.8	21.1
	2%	DB	30.1	33.8	36.2	38.1	35.9	33.1	32.0	31.2	31.8	31.8	29.9	29.9
		MCWB	18.6	18.2	19.0	20.8	22.8	24.0	24.3	24.4	24.5	23.6	20.8	20.7
	5%	DB	29.0	32.1	34.9	36.1	34.2	31.8	30.2	29.9	30.0	29.9	28.8	28.0
		MCWB	17.8	17.5	18.7	21.1	22.0	23.8	22.9	23.4	23.2	22.2	20.5	19.4
	10%	DB	27.9	30.8	33.1	34.2	33.1	30.0	29.0	28.2	28.9	28.2	27.2	26.9
		MCWB	16.9	16.7	18.0	19.2	21.9	22.6	22.1	22.2	22.5	21.1	19.8	18.8
	0.4%	WB	24.2	24.2	26.1	27.2	27.2	26.5	26.1	26.2	27.0	26.6	26.2	25.8
		MCDB	25.2	26.6	28.0	29.3	30.4	28.9	28.5	28.0	28.9	28.2	27.5	26.7
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.0	22.8	24.5	25.7	26.4	25.8	25.2	25.6	25.9	25.8	25.1	24.1
		MCDB	25.3	25.3	27.9	29.5	29.4	28.6	28.0	28.1	28.4	28.0	26.4	25.9
	5%	WB	21.8	21.4	23.5	24.1	25.4	25.2	24.7	25.0	25.1	25.1	24.0	23.1
		MCDB	24.9	25.4	27.1	29.4	28.8	28.5	27.7	28.1	28.2	27.2	25.3	25.2
	10%	WB	20.3	20.2	22.3	23.0	24.7	24.6	24.1	24.2	24.2	24.1	22.7	21.9
		MCDB	24.3	24.9	26.6	28.6	28.6	28.3	27.3	27.1	26.7	26.0	24.3	23.9
	5% DB	MDBR	13.4	15.0	14.1	12.5	10.5	8.1	7.5	7.2	7.8	8.7	9.0	10.4
		MCDBR	13.9	15.8	14.9	13.2	11.5	9.3	8.7	8.4	9.0	9.8	10.6	11.8
		MCWBR	4.4	4.6	5.0	5.8	3.8	2.6	2.3	2.3	2.7	3.3	3.6	4.1
		MCWBR	11.2	12.9	12.0	12.3	10.2	8.5	7.9	7.9	8.2	8.4	7.3	8.8
(42) Clear-Sky Solar Irradiance	taub		0.382	0.403	0.466	0.509	0.526	0.458	0.469	0.457	0.444	0.474	0.412	0.390
		taud	2.306	2.247	2.076	2.001	1.984	2.230	2.195	2.237	2.269	2.160	2.314	2.327
	Ebn at Noon		915	909	857	814	787	832	826	846	864	831	878	896
		Edh at Noon	127	140	170	183	183	141	147	143	138	151	126	122
(44) All-Sky Solar Radiation	RadAvg		5.38	6.16	6.53	6.59	6.29	5.29	4.71	4.77	5.14	4.96	4.62	4.71
	RadStd		0.24	0.25	0.23	0.22	0.29	0.38	0.29	0.28	0.28	0.37	0.42	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.32	N/A	N/A	+0.52	N/A	N/A	N/A	+171	+172

Nomenclature: See separate page